```
public class Hanoi {
   static int step = 1;
   static String A = "1,2,3", B = "-", C = "-";
   static void move(char from, char to) {
      String disk = "";
      if(from=='A') { disk=A.split(",")[0]; A=A.substring(disk.length()).replaceFirst("^,",""); }
      if(from=='B') { disk=B.split(",")[0]; B=B.substring(disk.length()).replaceFirst("^,",""); }
      if(from=='C') { disk=C.split(",")[0]; C=C.substring(disk.length()).replaceFirst("^,",""); }
      if(to=='A') A=disk+(A.equals("-")?"":","+A);
      if(to=='B') B=disk+(B.equals("-")?"":","+B);
      if(to=='C') C=disk+(C.equals("-")?"":","+C);
      System.out.println("\nStep "+step+++": "+from+" -> "+to);
      System.out.println("A: "+A+"\nB: "+B+"\nC: "+C);
   }
   static void solve(int n, char a, char b, char c) {
      if(n==1) { move(a,c); return; }
      solve(n-1,a,c,b);
      move(a,c);
      solve(n-1,b,a,c);
   }
   public static void main(String[] args) {
      System.out.println("Initial: A=1,2,3  B=-  C=-");
      solve(3,'A','B','C');
   }}
```

**Practical 3: Tower of Hanoi Using State Space Search**

**1. Theory**

State-space search is a problem-solving method in which every possible configuration is represented as a state.

In Tower of Hanoi, we have 3 rods and different-sized disks. The aim is to move all disks from the source rod to the destination rod using the auxiliary rod.

Names of the rods:

1. Source Rod (A): Starting rod where all disks are placed.
2. Auxiliary Rod (B): Helper rod used to move disks.
3. Destination Rod (C): Final rod where all disks must be placed.

Rules:

- Move only one disk at a time.
- Only the top disk can be moved.
- Never place a larger disk on a smaller disk.

For n disks, the minimum number of moves is:

\[ 2^n - 1 \]

For 3 disks, minimum moves = 7.

**2. Algorithm**

1. Start.
2. Create the initial state with all disks on source rod A.
3. Set destination rod C as the goal.
4. Insert the initial state into a queue.
5. Remove a state from the queue.
6. Check whether it is the goal state.
7. Generate all possible legal moves between rods.
8. Mark new states as visited and add them to the queue.
9. Repeat until the goal state is reached.
10. Display the moves and stop.

**3. Viva Questions and Answers**

Q1. What is state-space search?
It is a method of solving problems by searching through possible states.

Q2. What are the three rods called?
Source, Auxiliary and Destination.

Q3. What is the initial state?
All disks are placed on source rod A.

Q4. What is the goal state?
All disks are placed on destination rod C.

Q5. How many rods are used?
Three rods.

Q6. What are the rules of Tower of Hanoi?
Move one disk at a time and never place a larger disk on a smaller disk.

Q7. What is an operator?
An action used to move a disk from one rod to another.

Q8. What is path cost?
The total number of moves.

Q9. What is the minimum number of moves for n disks?
$2^n - 1$.

Q10. How many moves are required for 3 disks?
7 moves.

Q11. What is BFS?
Breadth-First Search explores states level by level.

Q12. Why is a visited list used?
To avoid visiting the same state again.

Q13. What is a legal move?
Moving a smaller disk onto a larger disk or an empty rod.

Q14. Why does BFS find the shortest solution?
Because every move has equal cost.

Q15. What is a state?
The arrangement of disks on the three rods.